Wu Hanju

Hong Kong, China | wuhanju@connect.hku.hk | hanju-wu.github.io

Research Interests

Mathematical optimization.

Education

Ph.D. in Mathematics 2023–present
The University of Hong Kong

B.Sc. in Mathematics 2019–2023
Sun Yat-sen University

- CGPA: 4.1/5.0 (91/100).
- Major GPA: 4.3/5.0 (93/100).
- Thesis: Complexity of a two-metric projection method for bound-constrained optimization.

Publications

Conference paper

- **Hanju Wu** and Yue Xie. "A study on two-metric projection methods." *2024 INFORMS Optimization Society Conference*. [arXiv:2409.05321] [IOS 2024 (Submission #5)]

Preprint

- **Hanju Wu** and Yue Xie. "On the resolution of ℓ_1 norm minimization via a two-metric adaptive projection method." [arXiv:2504.12260]
- **Hanju Wu**, Ching-pei Lee, and Yue Xie. "Transferability of Error Bounds and the Kurdyka-Łojasiewicz Property under C^1 Partial Smoothness." [arXiv:2605.02754] [Optimization Online]

Talks

On the resolution of ℓ_1 norm minimization via a two-metric adaptive projection method 2025.04
2025 East Coast Optimization Meeting, George Mason University, Arlington, Virginia, USA

On the resolution of ℓ_1 norm minimization via a two-metric adaptive projection method 2025.11
2025 APORS, HK Polytechnic University, Hong Kong, China

Transferability of Error Bounds and the Kurdyka-Łojasiewicz Property under C^1 Partial Smoothness 2026.11
2026 APORS (Session Chair), Nanyang Technological University, Singapore

Work Experience

Research Intern 2022.07–2022.08
BICMR, Peking University

Research Intern 2026.08–present
Theoretical Dept-Leibniz Research Center, 2012 laboratories, Huawei HK Research Center

Teaching Experience

MATH2012 Fundamental Concepts of Mathematics Spring 2026
The University of Hong Kong, Teaching Assistant

MATH3601 Numerical Analysis Spring 2025 and Fall 2025
The University of Hong Kong, Teaching Assistant

MATH1011 University Mathematics I Fall 2024
The University of Hong Kong, Teaching Assistant

DATA8008 Scalable Optimization Methods in Data Science — Spring 2024
The University of Hong Kong, Teaching Assistant

Honors and Awards

First-Class Scholarship — Sun Yat-sen University
Postgraduate Scholarship — The University of Hong Kong

Skills

Languages: Mandarin, Cantonese, English.
Programming: C, C++, Python (PyTorch), MATLAB, Julia, Lean.
Writing: LaTeX.